

Lab Experiment-04

4. Implement histogram equalization on the given image and compare it with the original image using Open CV.

Code:

```
!pip install opencv-python
import cv2
from google.colab.patches import cv2_imshow
from google.colab import files
uploaded = files.upload()
filename = list(uploaded.keys())[0]
img = cv2.imread(filename)
if img is None:
    print("Error: Image not loaded")
else:
    print("Image loaded successfully")
    cv2_imshow(img)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    equalized = cv2.equalizeHist(gray)
    print("Grayscale Image:")
    cv2_imshow(gray)
    print("Histogram Equalized Image:")
    cv2_imshow(equalized)
```

Output:



Grayscale Image:



Histogram Equalized Image:

